

Technical Document #535: Data Engineering - Comprehensive Overview

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Data pipelines automate the movement and transformation of data between systems. Apache Airflow and Prefect are popular workflow orchestration tools.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Remote work technologies have transformed the workplace. Collaboration tools and cloud services enable distributed teams to work effectively from anywhere.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Key Takeaways:

- Data lakes store raw data in its native format.
- Data warehouses store structured data optimized for analytical queries.
- Stream processing handles continuous data streams in real-time.